

Past-Papers-2016Short

what is supervised learning?

what is unsupervised learning?

what is knowledge?

what is support in data mining?

what do you mean by confidence in data mining?

what is posterior probability?

what is entropy?

what do you mean by nominal data?

what is the time consumed by k-mean algorithm?

$$O(I \times N \times K \times D)$$

iterations  
 (no of data points) (number of clusters) (dimensionality of data)

Give example of ordinal data?

- o Economic status (lower class, upper class, higher class)
- o Rating scale
- o Likert scales (agree, disagree)



DATE: 1/12/20

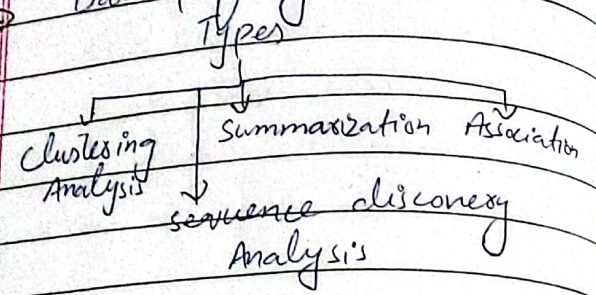
MTWTFSS

What is descriptive data mining?

→ main goal is to summarize or turn given data into relevant information.

→ identify patterns and  $\sigma$ s in data.

→ Data profiling



Name any technique that is more efficient than apriori?

- o Frequent pattern growth method.

Discuss disadvantages of k-mean algorithm?

- o Sensitive to outliers
- o Sensitive to initial centroid selection

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Assume spherical clusters of equal size

o Require predefined numbers of clusters

o may converge to local optima

o Not suitable for non-linear data.

o Does not incorporate noise handling.

Q. How do you find polarity for nominal data?

→ Polarity refers to +ive or -ive sentiment associated with textual or categorical data.

$$\text{polarity (+ve)} = \frac{\text{no of +ive instances}}{\text{no of instances in category}} \times 100$$

methods:

- frequency-based polarity
- Domain specific knowledge



Give example of ordinal data  
what do you mean by maximal  
in maximal frequent pattern  
growth mining?

⇒ A maximal pattern is a set  
of items that occurs frequently in  
the data, and there is no larger  
set of items with the same frequency.  
⇒ it's "maximal" because it  
cannot be expanded further  
without losing its frequency  
property.

How is k-medoids algorithm is  
better than k-means?

- o Robust to outliers
- o can handle clusters of unequal  
size
- o work with arbitrary distance  
metrics.
- o Handle non-numeric data.
- o Scalability
- o Deterministic results

Give an example of noise and  
inconsistent data?

- o noise can distort patterns & its in data.
- o Inconsistent can lead to error and anomalies  
in analysis outcome

noise: a score of 1000 out of 100, is outlier  
or data entry error.

inconsistent: entering -ive value in age or > 150 years.

what happen if we train the  
machine with false data?

- o can mess up learning process
- o wastage of resources & time.
- o making biased predictions/classification
- o leading to wrong decisions.
- o Poor performance

⇒ To avoid these problem:

- o Clean data
- o accurate data
- o Testing the model



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M T W T F S

Is privacy of data an issue in data mining? Give reason?

⇒ Yes, privacy of data is significant issue in data mining.

- can lead to identity theft or discrimination.
- Data breaches
- inference attack

⇒ To avoid this:

- implement data protection measures
- Privacy-preserving data mining methods.